

Module 2: Lines and Parabolas – Prequiz

Name:

Date:

Four short questions. Show the algebra that supports each answer.

1. Find the slope of the line through $(-1, 2)$ and $(3, 10)$.

2. Find an equation of the line with slope -3 that passes through $(2, 5)$.

3. For

$$y = 2(x + 4)^2 - 1,$$

state the vertex and tell whether the parabola opens upward or downward.

4. Complete the square to rewrite

$$x^2 + 6x + 1$$

in vertex form.

Module 2: Lines and Parabolas – Weekly Check-In

Name:

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Three questions. Show the algebra that supports each answer.

1. A line from two points.

A line passes through $(-2, 1)$ and $(4, 10)$.

- (a) Find the slope.

- (b) Find an equation of the line in slope-intercept form.

2. Completing the square.

Consider

$$q(x) = x^2 - 6x + 5.$$

Complete the square to rewrite $q(x)$ in vertex form. Then state the vertex and whether the parabola opens upward or downward.

Module 2: Lines and Parabolas – Weekly Check-In, continued

Name:

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3. Transforming and graphing a parabola.

Begin with the parent function $f(x) = x^2$ and define

$$g(x) = -2f(x - 1) + 3.$$

- (a) Describe the transformations from f to g in words.
- (b) Write $g(x)$ without function notation and state its vertex.
- (c) Graph $g(x)$ below. Clearly mark the vertex and at least two symmetric pairs of points.

